Diffusion-Based Handwritten Image Deblurring and Restoration

Dataset: Koushik Handwritten Character/Text Dataset (HCCD) Model: U-Net with VGG16 Perceptual Loss Final Resolution: 512x512

Course code-CSE-414

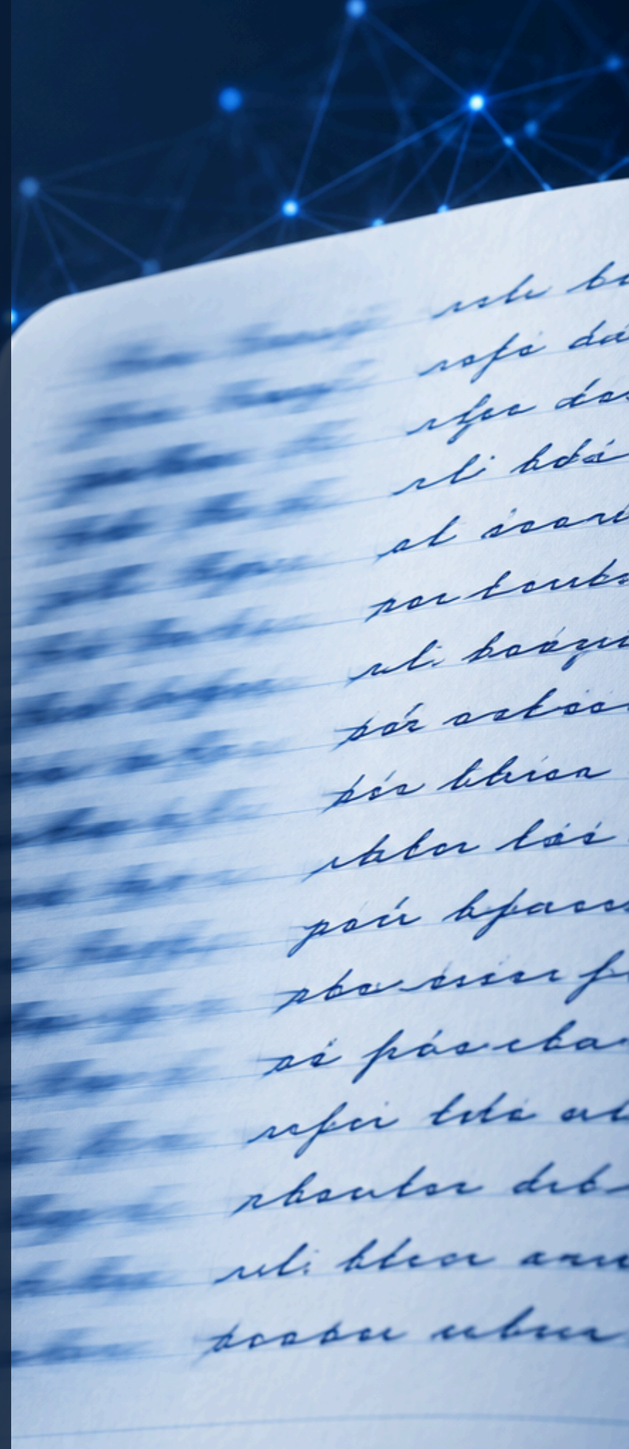
Course title- Machine learning and deep learning Lab



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Introduction

Handwritten documents often become blurry due to camera shake, poor focus, or scanning errors, making them difficult to read and process through OCR. Our project aims to restore blurry English and Bangla handwriting using an AI-based model, helping preserve historical documents and improve digital accessibility.



To solve this, we propose:

- A **U-Net-based deblurring system** trained on the **Koushik HCCD dataset** (1,722 pairs).
- A **Composite VGG16 Perceptual Loss** at **512×512 resolution**.
- Deployment via a **Gradio Web Interface** for real-time restoration.

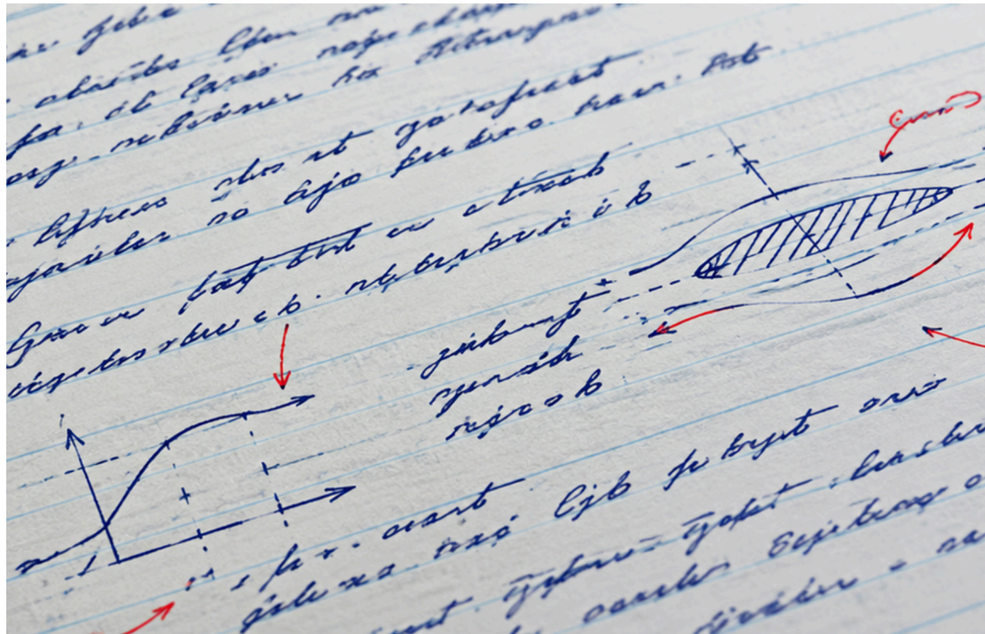
Problem Definition & Solution

- **Unreadable Text:** Blurry images lose crucial high-frequency details (pen strokes).
- **Filter Limitations:** Traditional sharpening filters cannot recover already lost information.
- **Over-Smoothing:** Basic Deep Learning models (like L1 loss) produce artificial, blurry outputs.



Solution -

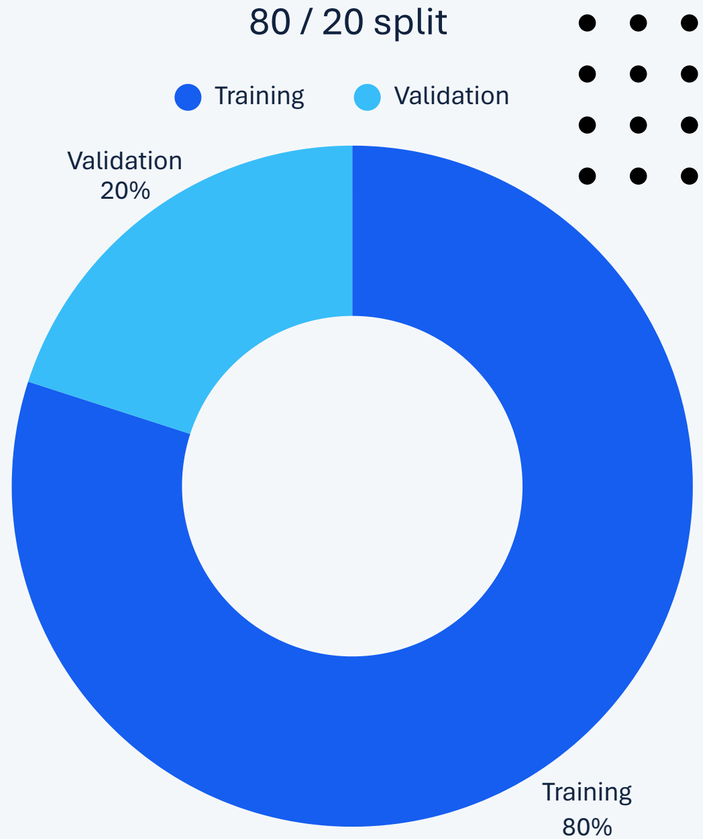
- **L1 Loss:** Ensures pixel-level accuracy.
- **SSIM Loss:** Preserves structural similarity and prevents text distortion.
- **VGG16 Perceptual Loss:** Forces the model to generate text with natural sharpness, mimicking how the human eye perceives detail.



Dataset & Preprocessing

Dataset: Koushik HCCD

We used the **HCCD English Dataset**, containing **574 ground-truth** images with three blur levels, generating a total of 1,722 image pairs. For **Bangla,** we collected **176 clear** handwriting images and applied **Gaussian Blur** using **OpenCV** to create synthetic blurry images. Finally, both datasets were combined to create **1,898 image pairs,** divided into **80% training** and **20% validation.** All images were resized to **512×512 pixels** and normalized before training.



574×3 =1,722
Total=1898

System Architecture (U-Net)

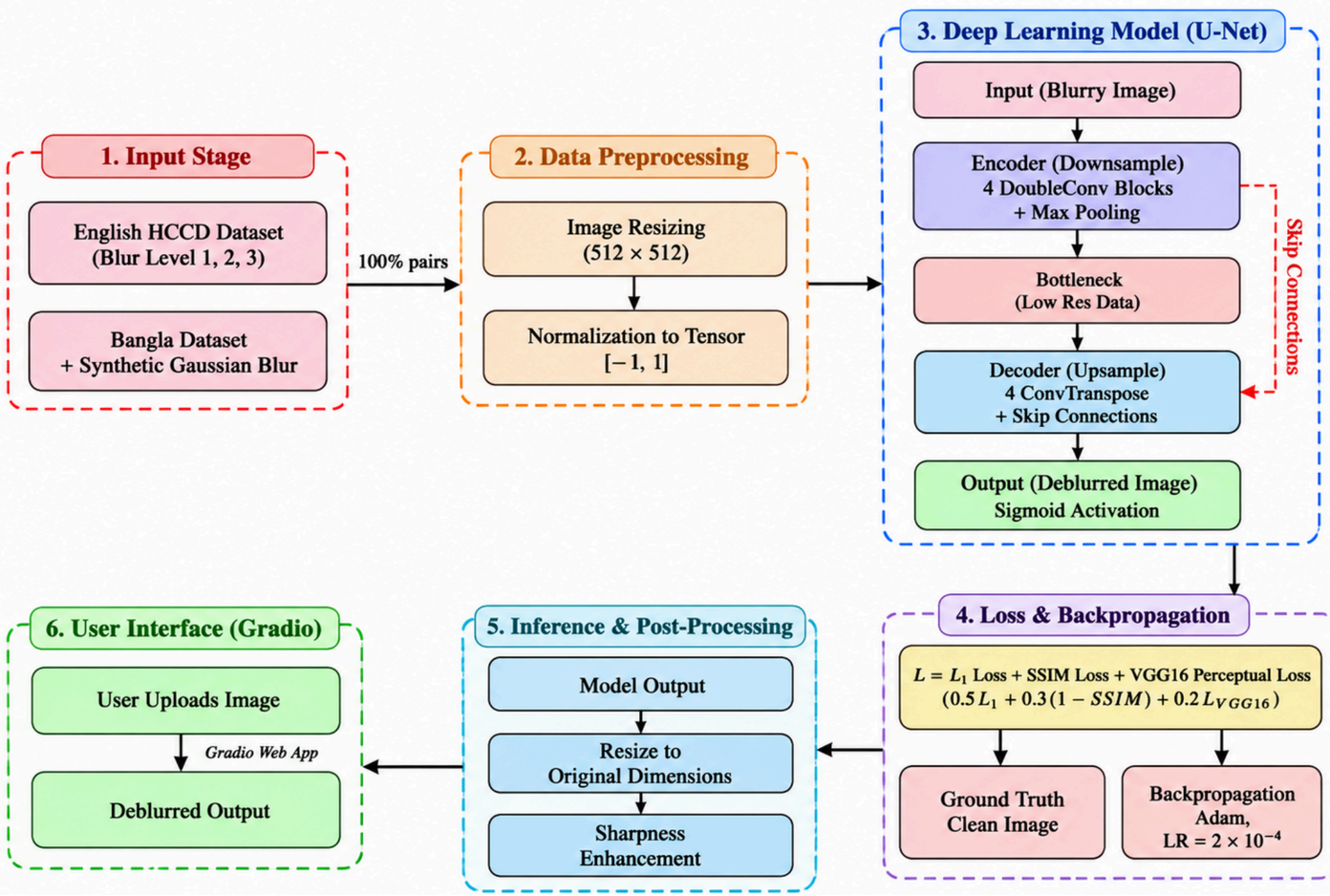
Encoder: 4× Downsampling (DoubleConv + MaxPool).

Bottleneck: Deepest feature representation.

Decoder: 4× Upsampling (ConvTranspose) + Skip Connections.

Output: Clean Image via Sigmoid activation (~7.7M parameters).

Visual: A clean U-Net diagram (down-sampling blocks, bottleneck, up-sampling blocks with skip connection arrows).



Methodology

1. Dataset Preparation (Building the Pairs)

To train the AI, we needed both "blurry" and "clear" images.

- **English (HCCD):** Used existing data (1,722 paired images).
- **Bangla:** Since no blurry Bangla images existed, we used **Gaussian Blur** (via OpenCV) to artificially blur 176 clear images.
- **Result:** A combined dataset of **1,898 image pairs**.

2. Model Architecture (The U-Net)

We used a **U-Net** deep learning model.

- **Why U-Net?** It has an "Encoder" to analyze the blur and a "Decoder" to rebuild the image.
- **Skip Connections:** The secret weapon of U-Net. It passes original details directly to the decoder, ensuring the text structure and fine lines are not lost.

Methodology

3. Loss Function (How the AI Learns)

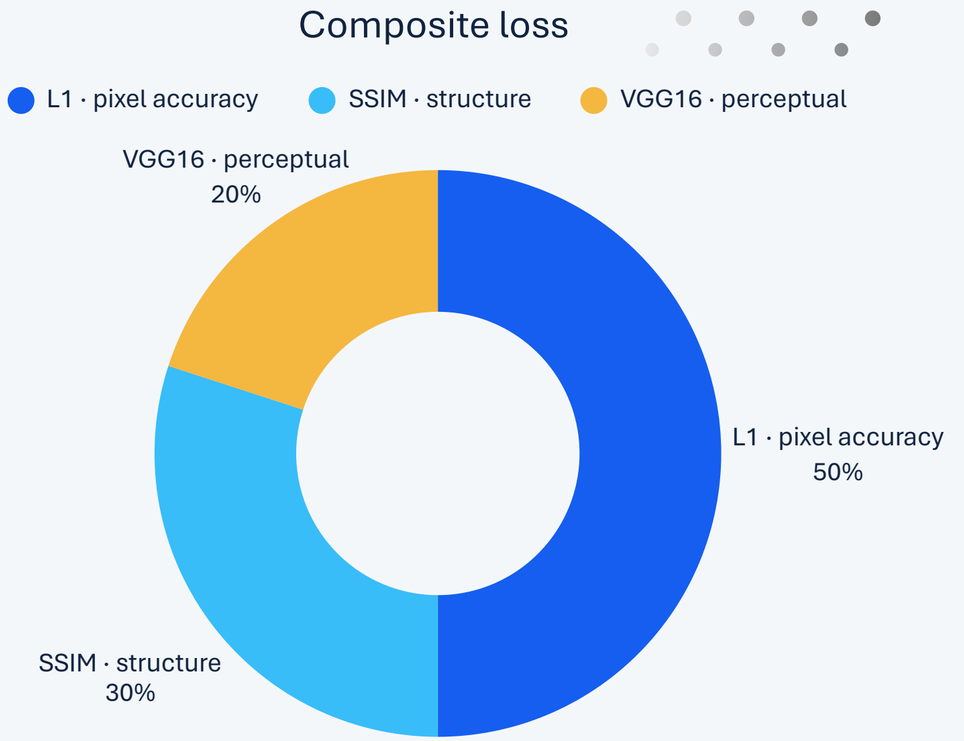
- We used a **Composite Loss Function** to get the sharpest result:
- **50% L1 Loss:** Fixes pixel-by-pixel errors.
- **30% SSIM Loss:** Maintains the structure of the letters.
- **20% VGG Perceptual Loss:** Makes the image look natural and sharp to the human eye.

4. Training Process

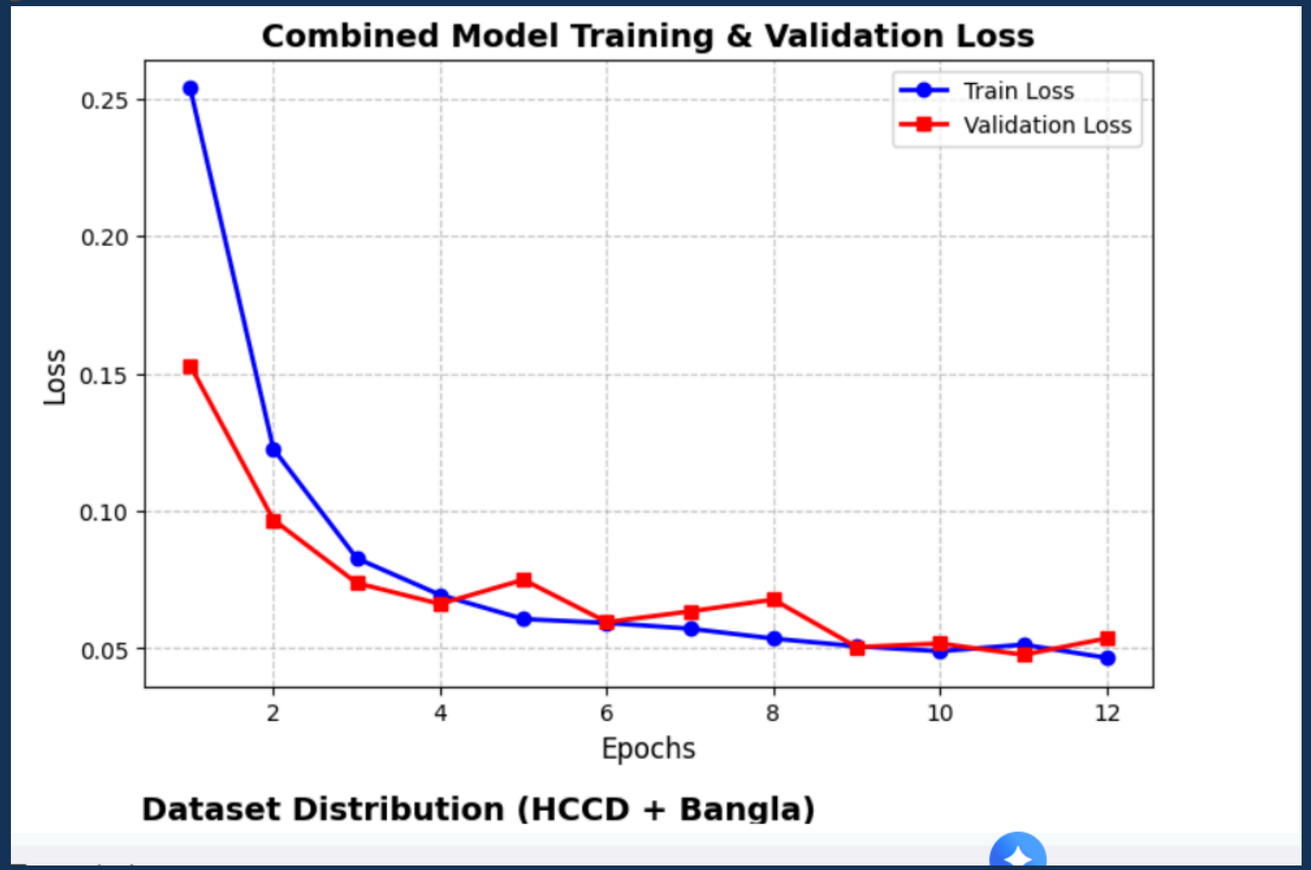
- Image Size: Resized to 512x512 pixels.
- Hardware: NVIDIA T4 GPU (Google Colab).
- Training: 12 Epochs using the Adam Optimizer (Learning Rate: 0.0002).
- Result: The model successfully learned to deblur both English and Bangla text, achieving a best validation loss of 0.0477.

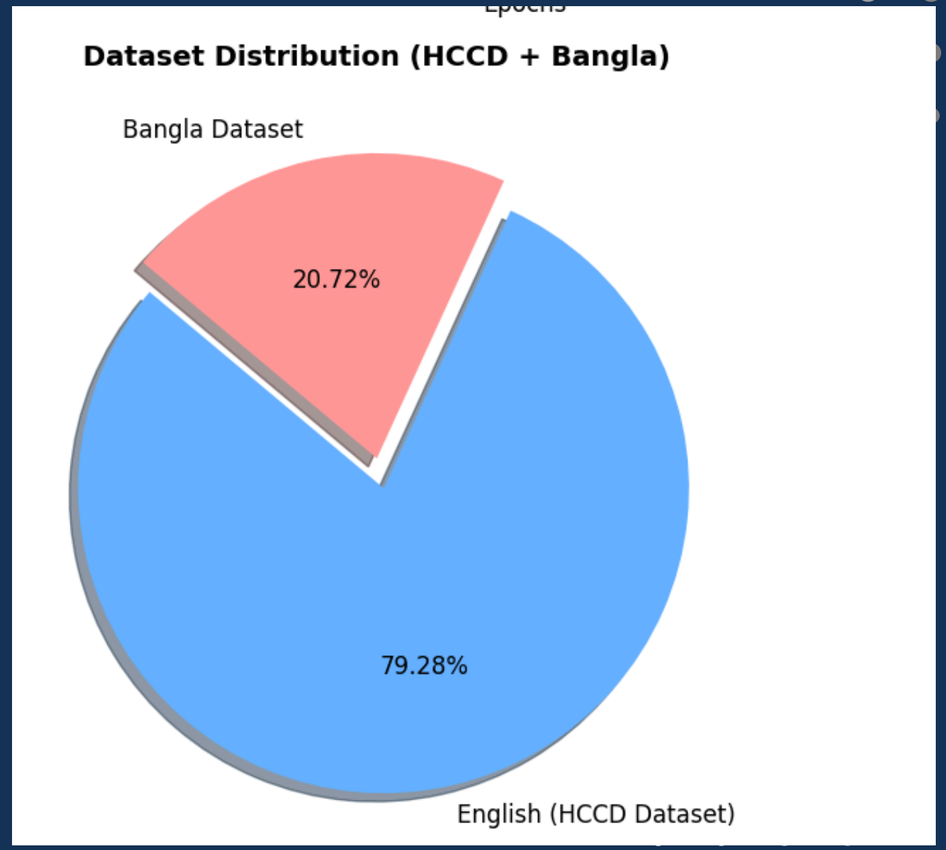
Loss Function & Training

To achieve optimal restoration, we trained the model using a **composite loss function** with a specific weight distribution. We allocated **50% to L1 Loss** to ensure exact pixel accuracy, **30% to SSIM Loss** for structural preservation, and **20% to VGG16 Perceptual Loss** to generate human-like sharpness. Additionally, we upgraded the training **resolution to 512×512** and utilized the **Adam optimizer** with a **learning rate of 3e-4** to effectively update the model weights.

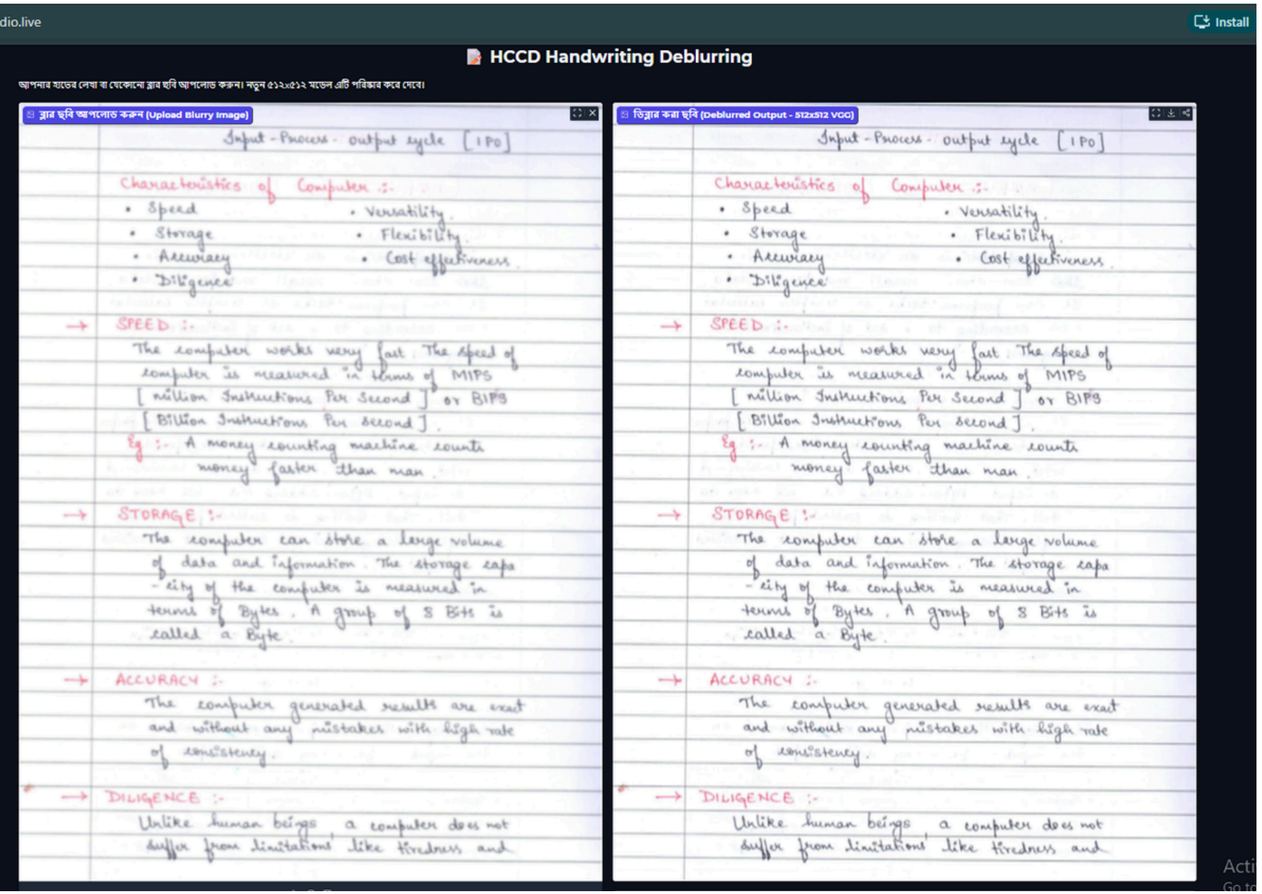


Model Train Diagram

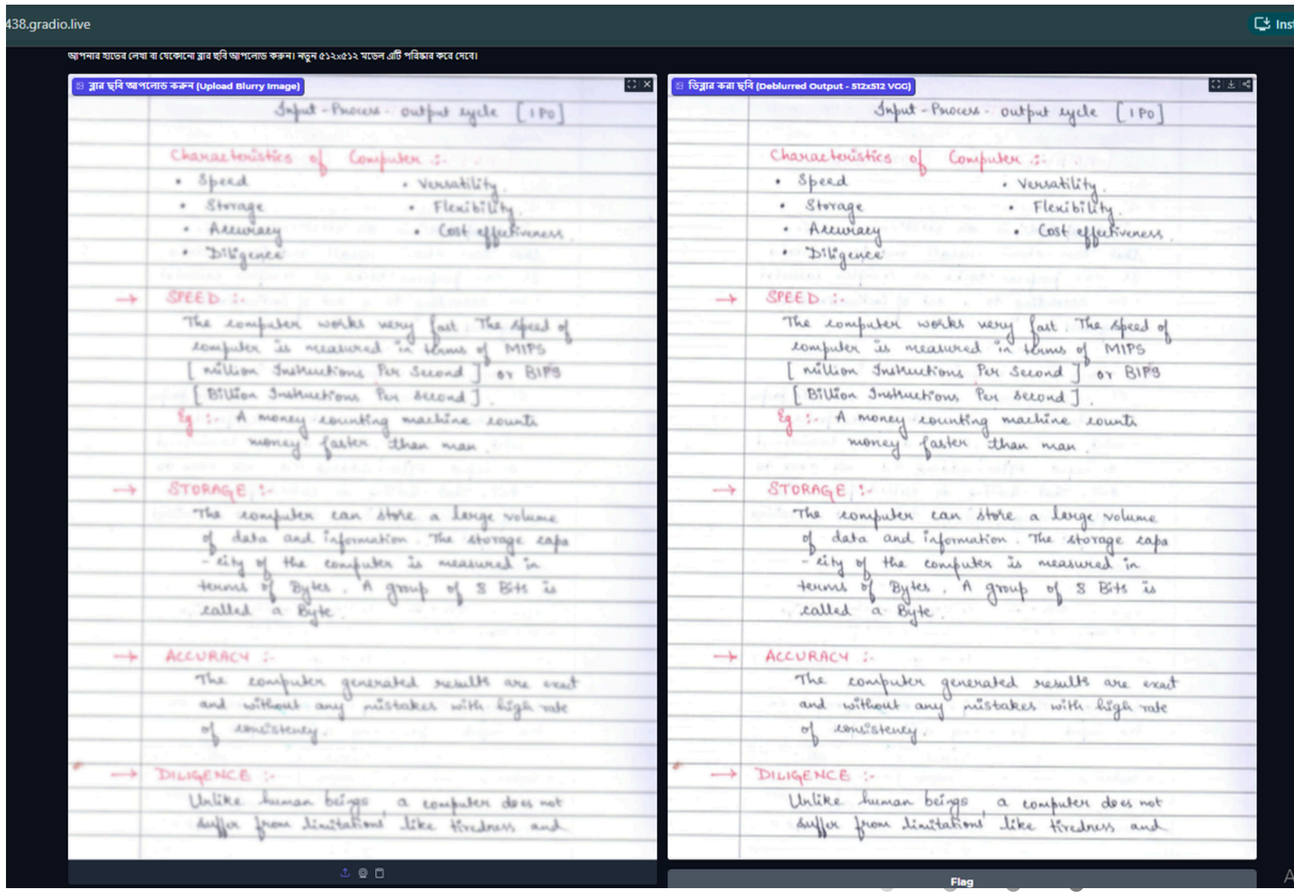




Results & Visual Output

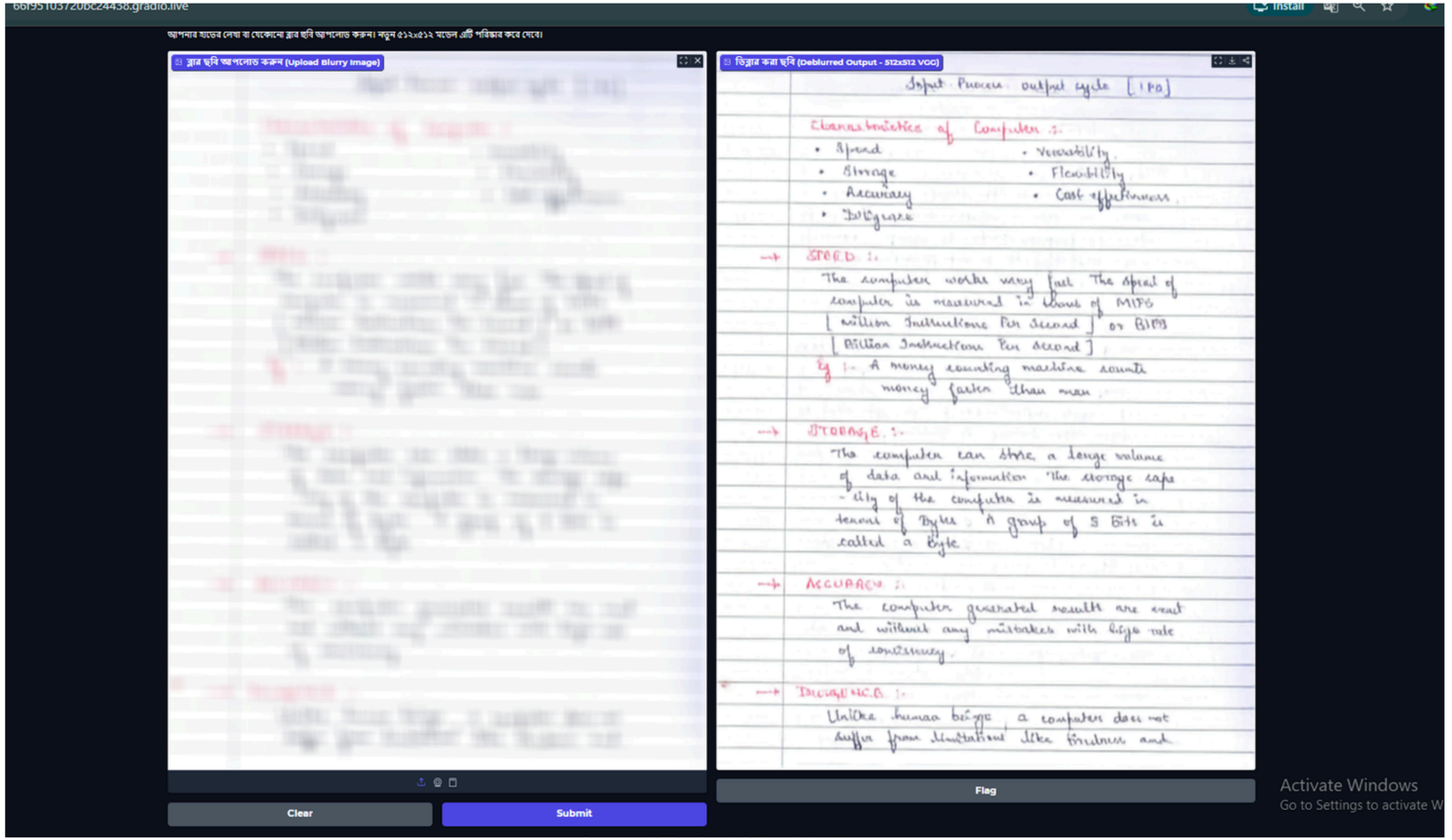


Level-1(English de-blur)



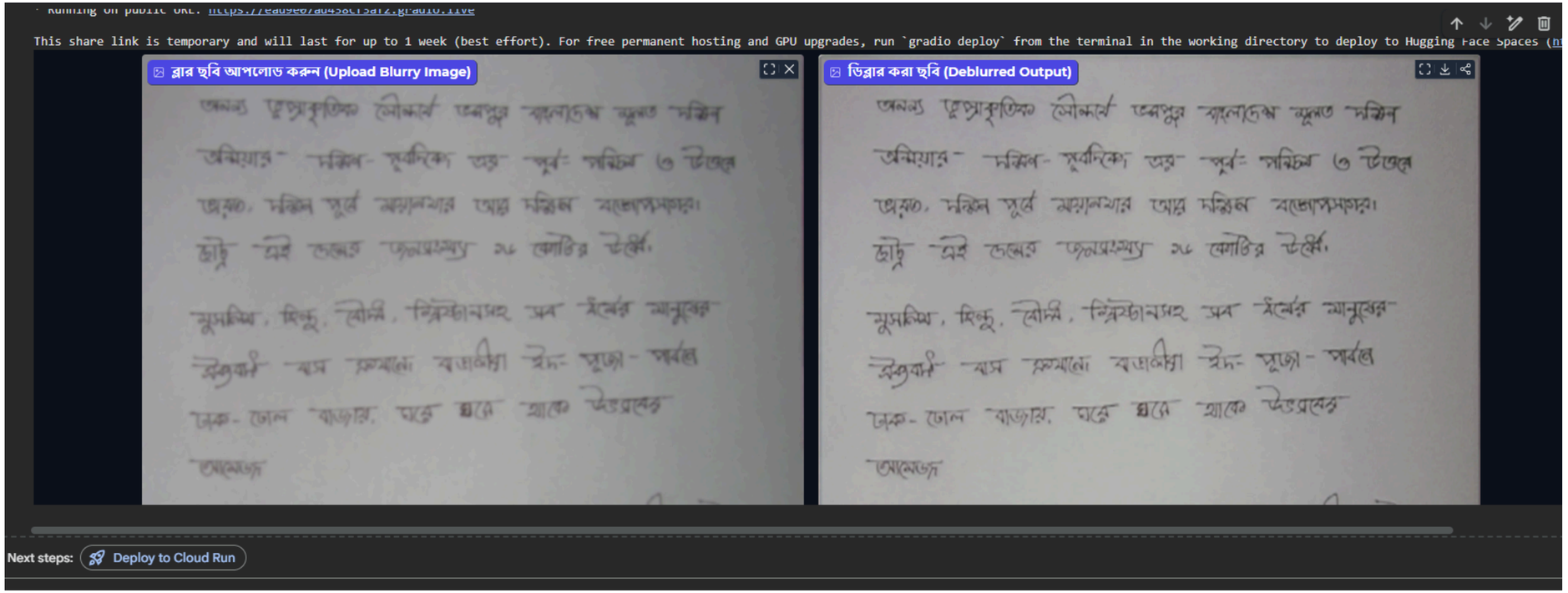
Level-2(English de-blur)

Results & Visual Output



Level-3(English de-blur)

Results & Visual Output



Bangla Blur to De-Blur

Baseline (256×256): PSNR = 35.89 dB, SSIM = 0.9653.

Final Output (512×512 + VGG):
Crystal clear, no over-smoothing, red ink preserved.

Contribution to SDGs

Our project aligns with three key United Nations Sustainable Development Goals. First, it supports **SDG 4 (Quality Education)** by providing **clear digitized notes** for students to study more effectively. Second, it contributes to **SDG 9 (Innovation)** by developing an **AI tool for digitizing historical records**, modernizing document preservation. Finally, it promotes **SDG 10 (Reduced Inequalities)** by enabling **good quality restoration even with low-end cameras**, making digital literacy accessible to everyone.



Limitations & Ethics

- **Synthetic Blur Only:** Trained exclusively on synthetic blurs; may struggle with real-world motion blur or severe out-of-focus artifacts.
- **Handwriting Bias:** Dataset is specific to a single handwriting style. Accuracy may drop on different cursive fonts or distinct writing styles.

- **Ethical Compliance:** No personal or sensitive data was used. All handwriting samples were completely anonymized and collected with consent.



Future Work

- Train at higher resolutions (1024x1024) for HD quality.
- Collect more real-world blurry Bangla documents.
- Deploy the Gradio app to Hugging Face Spaces for permanent access.

Our future roadmap includes training at resolution for HD results, building a mobile app for offline use, and integrating OCR to turn restored handwriting into editable text."

Conclusion

In conclusion, we successfully built a **U-Net model** that effectively restores sharp handwriting from blurred images. To make this technology accessible, we deployed the system via a **Gradio interface**, allowing anyone to use it easily. For future work, we plan to train the model on **real-world blur datasets** to improve its robustness and integrate it with **OCR** for automated text extraction.

References

[1] O. Ronneberger, P. Fischer, and T. Brox, “U-Net: Convolutional networks
for biomedical image segmentation,” in Proc. Med. Image Comput. Comput.-Assist. Intervent. (MICCAI), 2015, pp. 234–241.

[2] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, “Image quality assessment: From error visibility to structural similarity,” IEEE
Trans. Image Process., vol. 13, no. 4, pp. 600–612, Apr. 2004.

[3] J. Johnson, A. Alahi, and L. Fei-Fei, “Perceptual losses for real-time
style transfer and super-resolution,” in Proc. Eur. Conf. Comput. Vis. (ECCV), 2016, pp. 694–711.

[4] “Koushik HCCD dataset,” Google Drive Repository, 2024.

[5] “SAIPL Bangla handwriting dataset,” Google Drive Repository, 2024.



Thank you

